**Medicore – Assignment 2 – OOPT**

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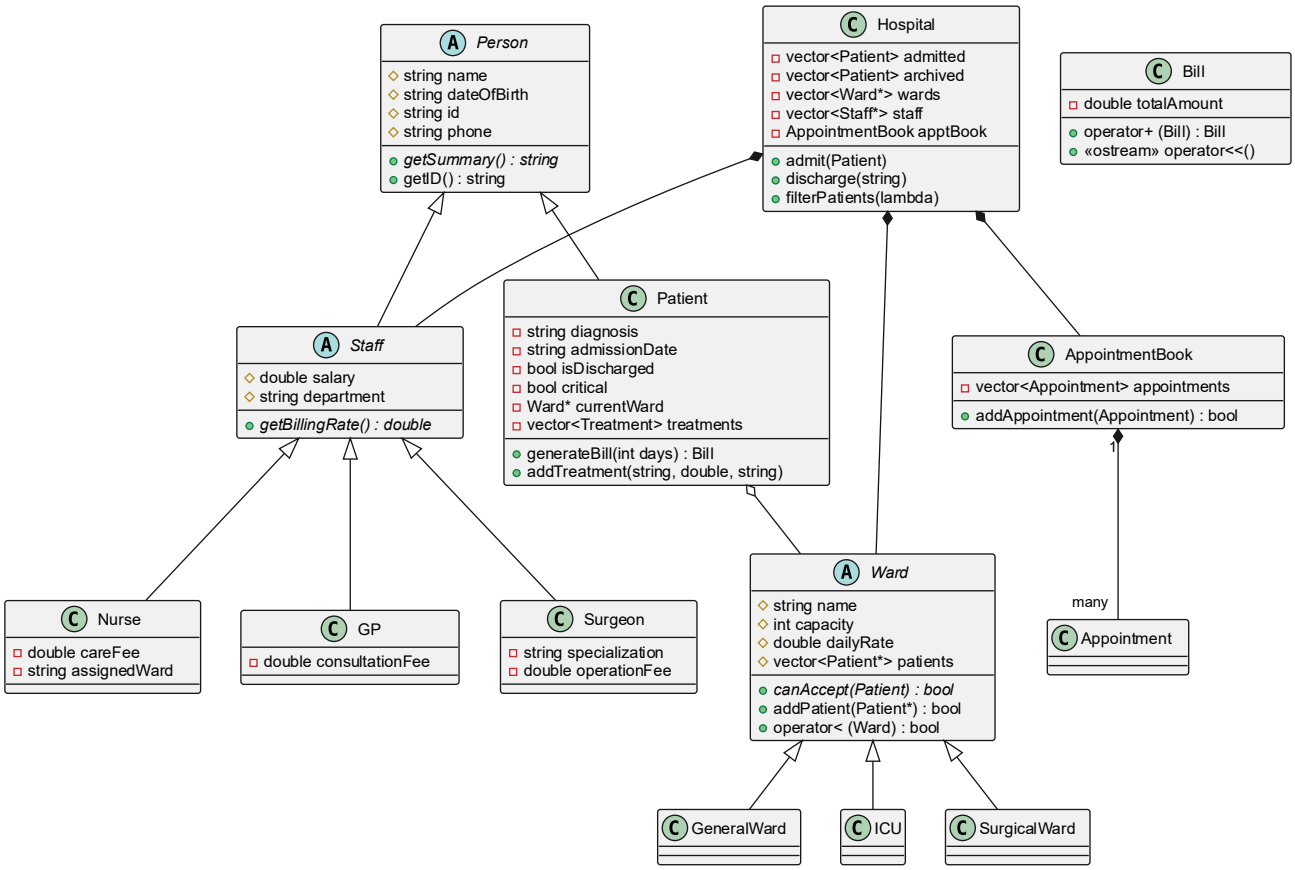
## Part1: UML Diagram:




## Part 2: Memory Ownership Diagram

| Resource Category | Owner (Responsible for delete) | Allocation Site | Lifetime Details |
|---|---|---|---|
| Ward Objects | Hospital Class | main.cpp (via new) | Created in main, passed to addWard(), deleted in Hospital::~Hospital(). |

| | | | |
|---|---|---|---|
| Staff Objects | Hospital Class | main.cpp (via new) | Created in main, passed to addStaff(), deleted in Hospital::~Hospital(). |
| Patient Objects | Hospital Class | Stack / vector<Patient> | Managed by std::vector value semantics. No manual delete needed. |
| Treatment Data | Patient Class | std::vector | Internally managed. Moves from admitted to archived via move semantics. |

## Part 3: Written Justification of Design Decisions

### 1. Template Method Pattern for Admissions in Ward

I implemented Ward::addPatient as a base function that calls the virtual canAccept. This ensures that both capacity constraints and specialized medical criteria are enforced. It prevents incorrect admissions, such as an ICU accepting stable patients, even when accessed via a Ward* pointer.

### 2. Move Semantics for Discharging a Patient

I used move constructors and move assignment operators for the Patient class. Since a patient's vector<Treatment> may be large, move semantics allow efficient transfer of treatment data from active to archived records in constant time without copying heap memory.

### 3. Used Pure Polymorphism instead of dynamic cast

I defined getBillingRate() as a pure virtual function in the Staff class. This avoids the need for dynamic_cast and leverages runtime polymorphism via the Virtual Table. The system directly calls the appropriate implementation based on the actual object type.

### 4. Lambda-Driven Reporting

Iused std::function and lambda expressions for filtering and sorting operations. This keeps the Hospital class independent of specific reporting logic and enables flexible, user-defined criteria without modifying the core system.